

EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Why is Tableau a preferred tool for data visualization?
Tableau is preferred because it provides an easy drag-and-drop interface, creates interactive dashboards, connects to multiple data sources, performs fast data analysis, and produces visually appealing charts without requiring programming knowledge.
2. What is the primary use of a line chart in data analysis?
A line chart is mainly used to display trends and changes in data over time. It helps identify patterns, growth, seasonal variations, and fluctuations.
3. When would you use a tree map instead of a bar chart?
A tree map is used when visualizing hierarchical data or comparing the proportion of many categories within a limited space. It is more suitable than a bar chart when there are numerous categories and sub-categories.
4. What is the significance of filters in Tableau visualizations?
Filters allow users to focus on specific data by selecting values such as year, region, category, or customer segment. They improve analysis by making dashboards interactive and easier to explore.
5. How does Tableau handle geographic data for visualizations?
Tableau automatically recognizes geographic fields such as Country, State, City, and Postal Code. It assigns latitude and longitude coordinates to generate interactive maps without requiring manual mapping.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

Tableau - Book1

File Data Server Window Help

Connections

Global_Superstore2

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Global_Superstore2.csv

New Union

New Table Extension

Go to Worksheet

Data Source

Sheet 1

Global_Superstore2

Global_Superstore2.csv

24 fields 51290 rows

100 rows

Name Global_Superstore2.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
#	Row ID	Global_Superstore2...	Row ID
Abc	Order ID	Global_Superstore2...	Order ...
Abc	Order Date	Global_Superstore2...	Order ...
Abc	Ship Date	Global_Superstore2...	Ship D...
Abc	Ship Mode	Global_Superstore2...	Ship ...
Abc	Customer ID	Global_Superstore2...	Custo...
Abc	Customer Name	Global_Superstore2...	Custo...
Abc	Segment	Global_Superstore2...	Segm...
Abc	City	Global_Superstore2...	City
Abc	State	Global_Superstore2...	State
Abc	Country	Global_Superstore2...	Country
Abc	Postal Code	Global_Superstore2...	Postal...

#	Global_Superstore2.csv	Abc	Global_Superstore2.csv	Global_Superstore2.csv	Global_Superstore2.csv	Abc	Global_Superstore2.csv	Abc	Global_Superstore2.csv	Abc	Global_Superstore2.csv
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	Country	Postal Code
32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495						
26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	JR-16210						
25330	IN-2013-71249	17-10-2013	18-10-2013	First Class	CR-12730						
13524	ES-2013-1579342	28-01-2013	30-01-2013	First Class	KM-16375						
47221	SG-2013-4320	05-11-2013	06-11-2013	Same Day	RH-9495						
22732	IN-2013-42360	28-06-2013	01-07-2013	Second Class	JM-15655						
30570	IN-2011-81826	07-11-2011	09-11-2011	First Class	TS-21340						
31192	IN-2012-86369	14-04-2012	18-04-2012	Standard Class	MB-18085						
40155	CA-2014-135909	14-10-2014	21-10-2014	Standard Class	JW-15220						
40936	CA-2012-116638	28-01-2012	31-01-2012	Second Class	JH-15985						
34577	CA-2011-102988	05-04-2011	09-04-2011	Second Class	GM-14695						
28879	ID-2012-28402	19-04-2012	22-04-2012	First Class	AJ-10780						
45794	SA-2011-1830	27-12-2011	29-12-2011	Second Class	MM-7260						
4132	MX-2012-130015	13-11-2012	13-11-2012	Same Day	VF-21715						
27704	IN-2013-73951	06-06-2013	08-06-2013	Second Class	PF-19120						

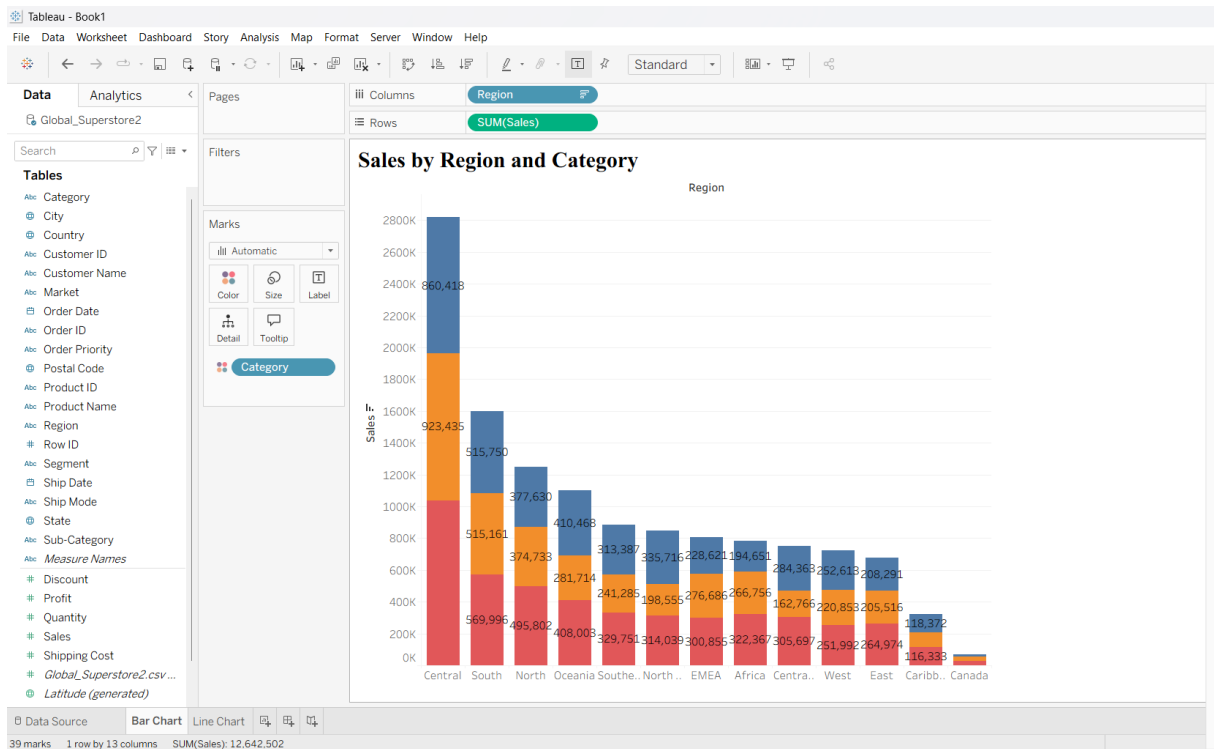
Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

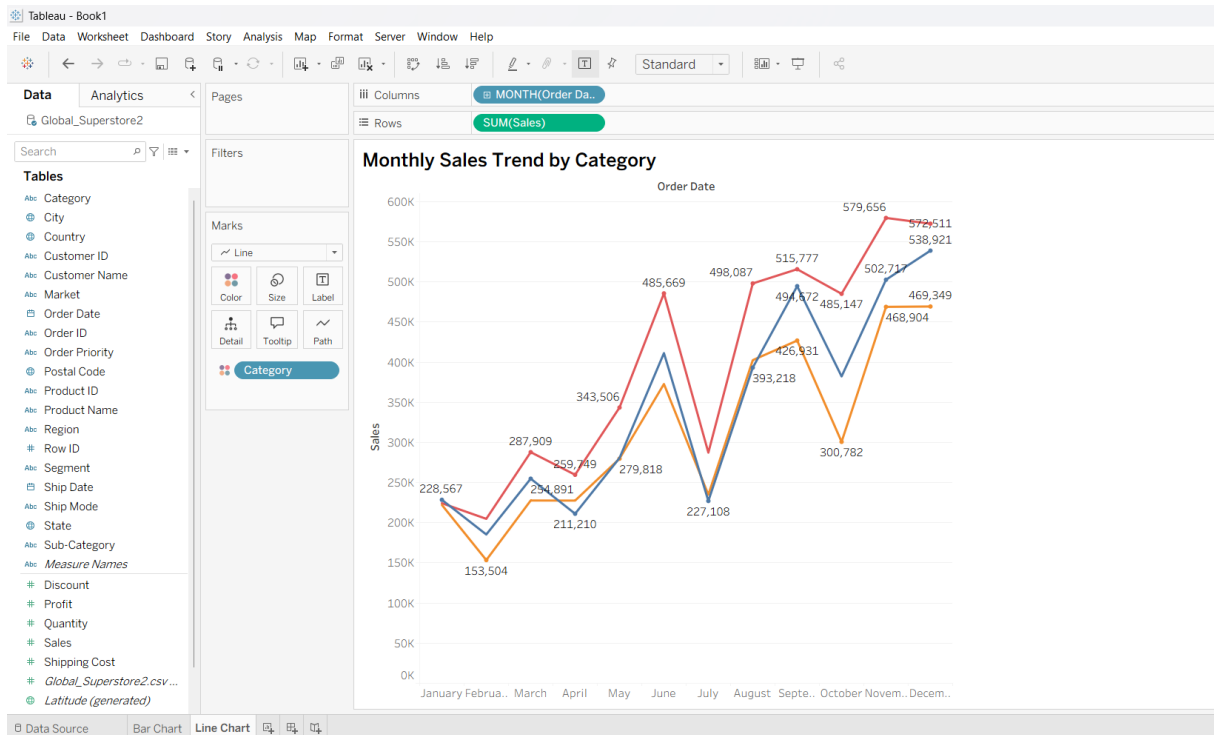


b) Line Chart:

Objective: Visualize sales trends over time.

Place Order Date on the x-axis and Sales on the y-axis.

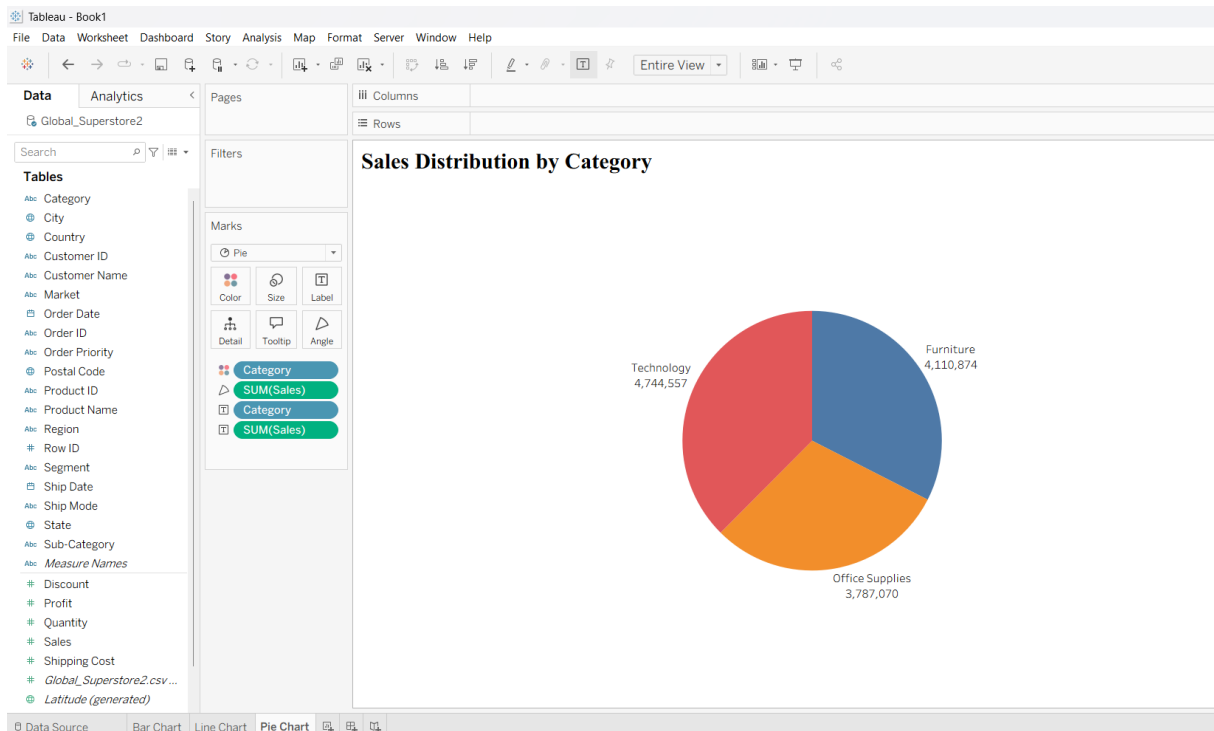
Set granularity to Month and color by Category.



c) Pie Chart:

Objective: Show sales distribution by product category.

Assign Category as the dimension and Sales as the measure.

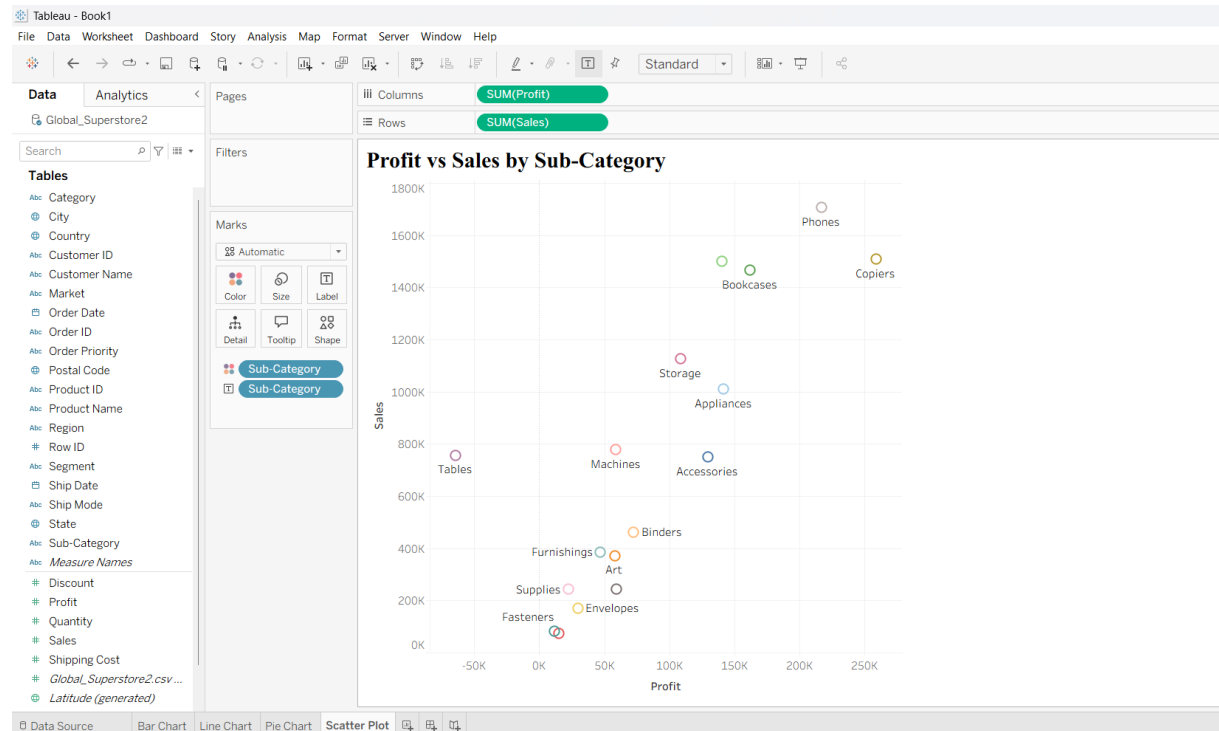


d) Scatter Plot:

Objective: Examine the relationship between profit and sales.

Plot Profit on the x-axis and Sales on the y-axis.

Use Sub-Category as the color dimension.

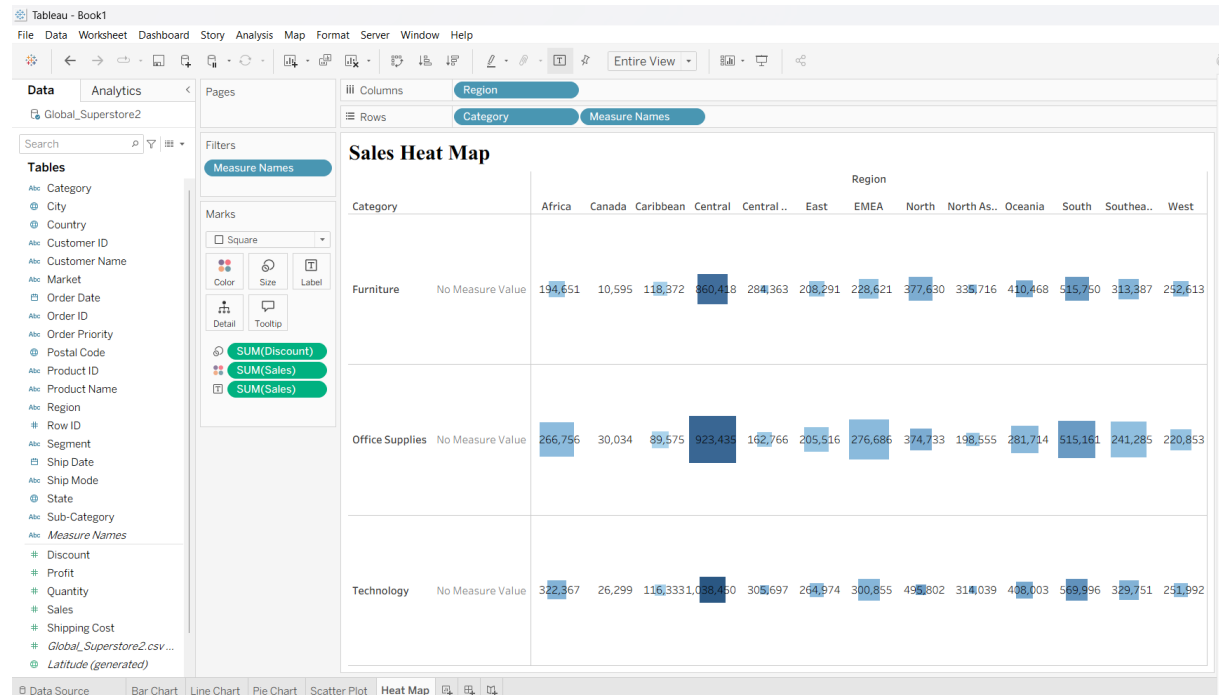


e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions.

Use Sales as the measure and apply color coding.

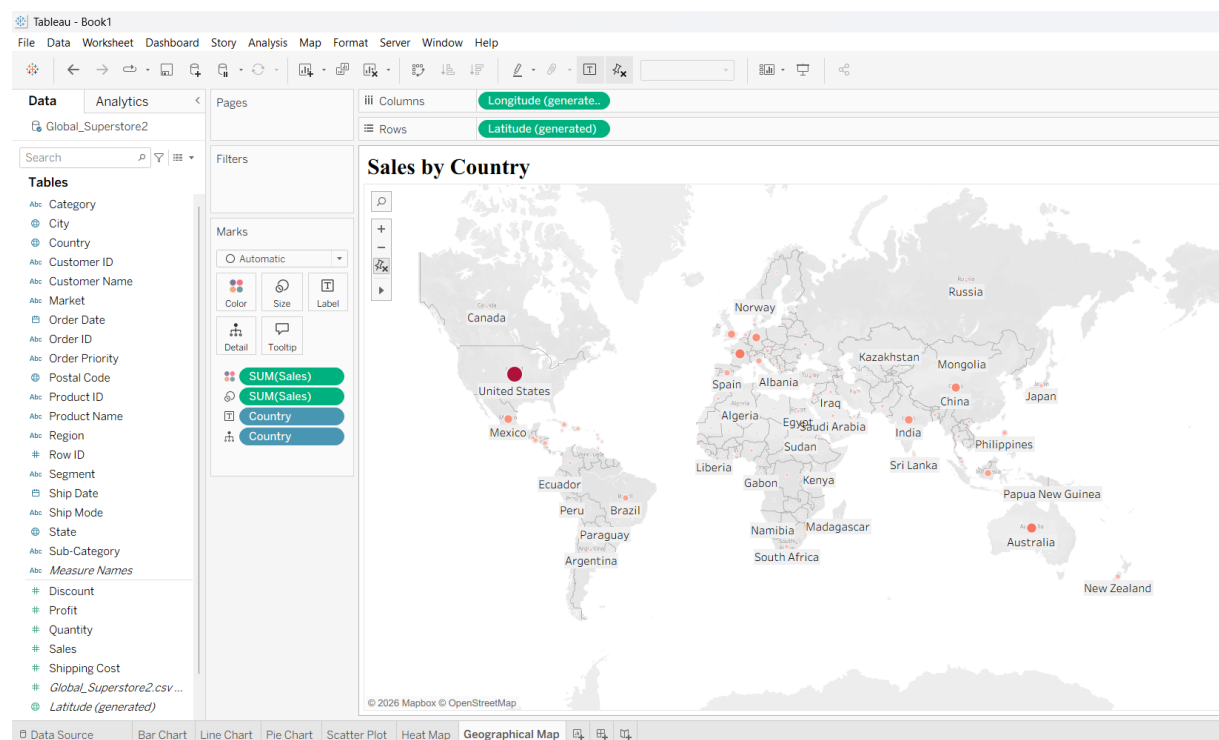


f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field.

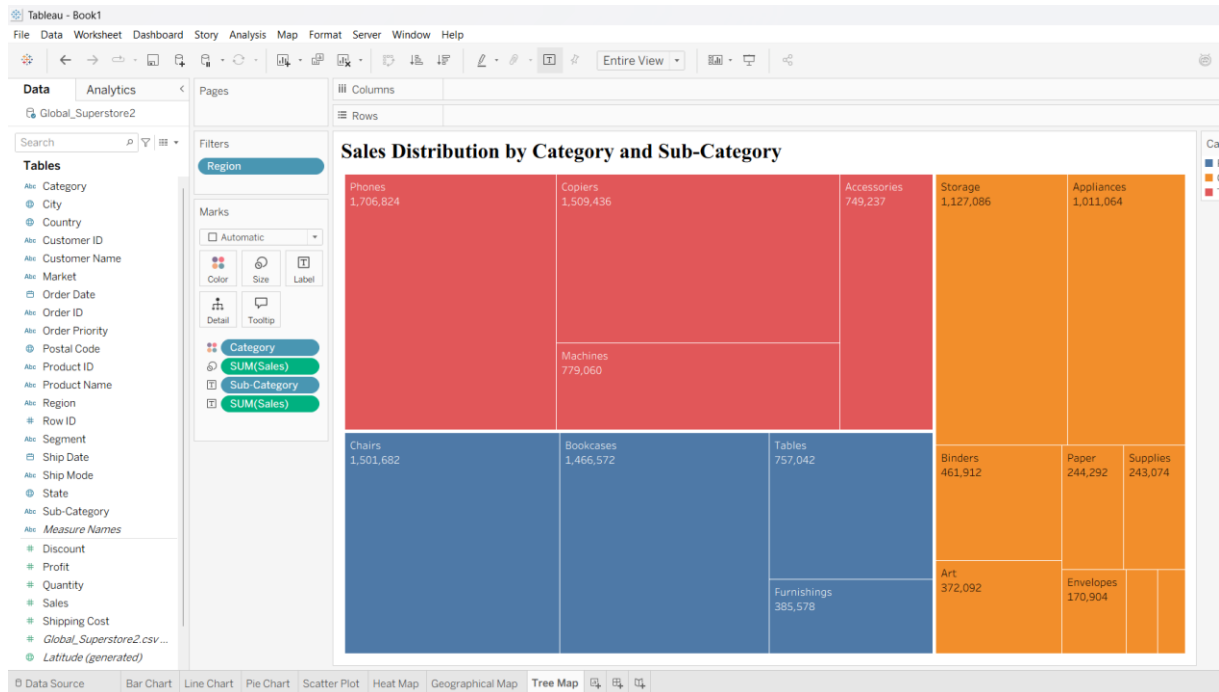
Use Sales to define size and color gradients.



g) Tree Map:

Objective: Display sales distribution by category and sub-category.

Use Category and Sub-Category as dimensions, and Sales as the measure.



Customization and Interactivity:

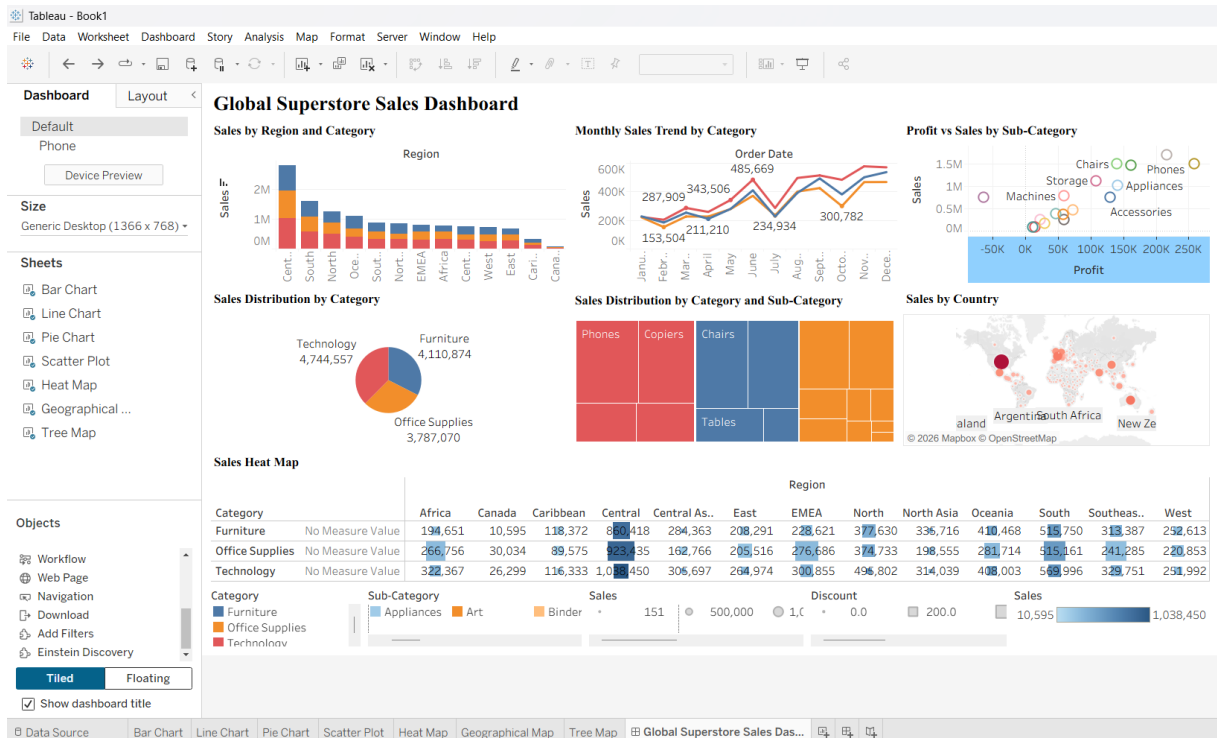
Add filters to explore data subsets interactively (e.g., by year, region, or category).

Include annotations, labels, and formatting for clarity.

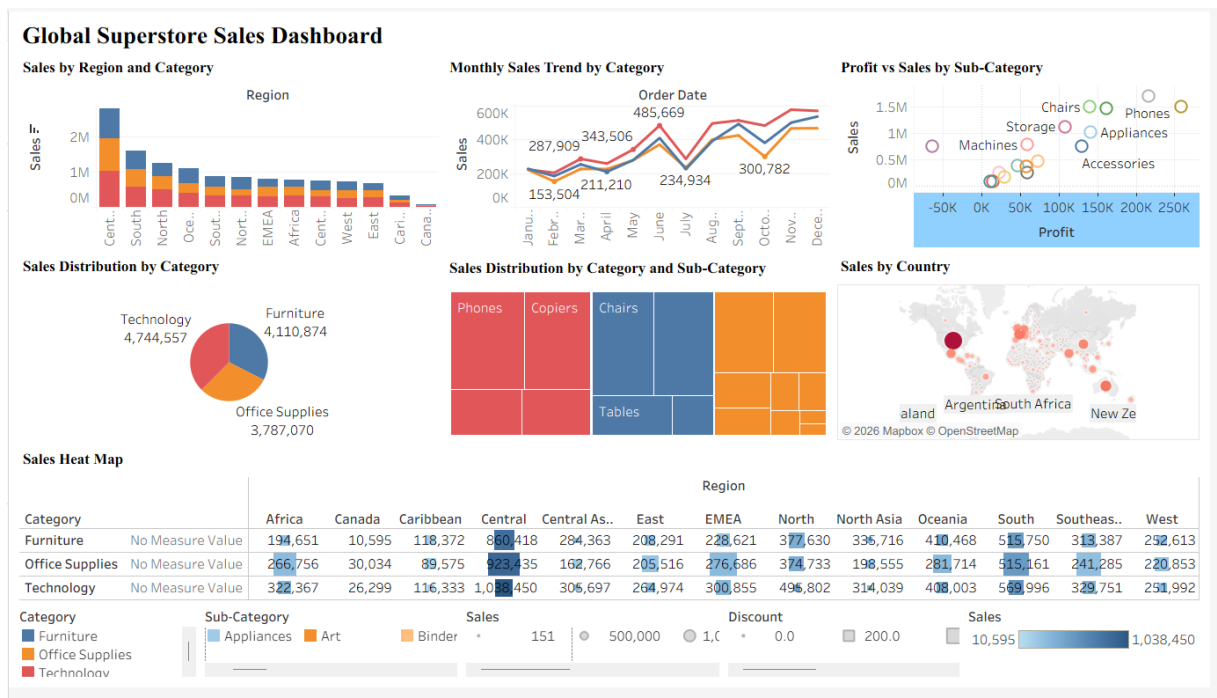
Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

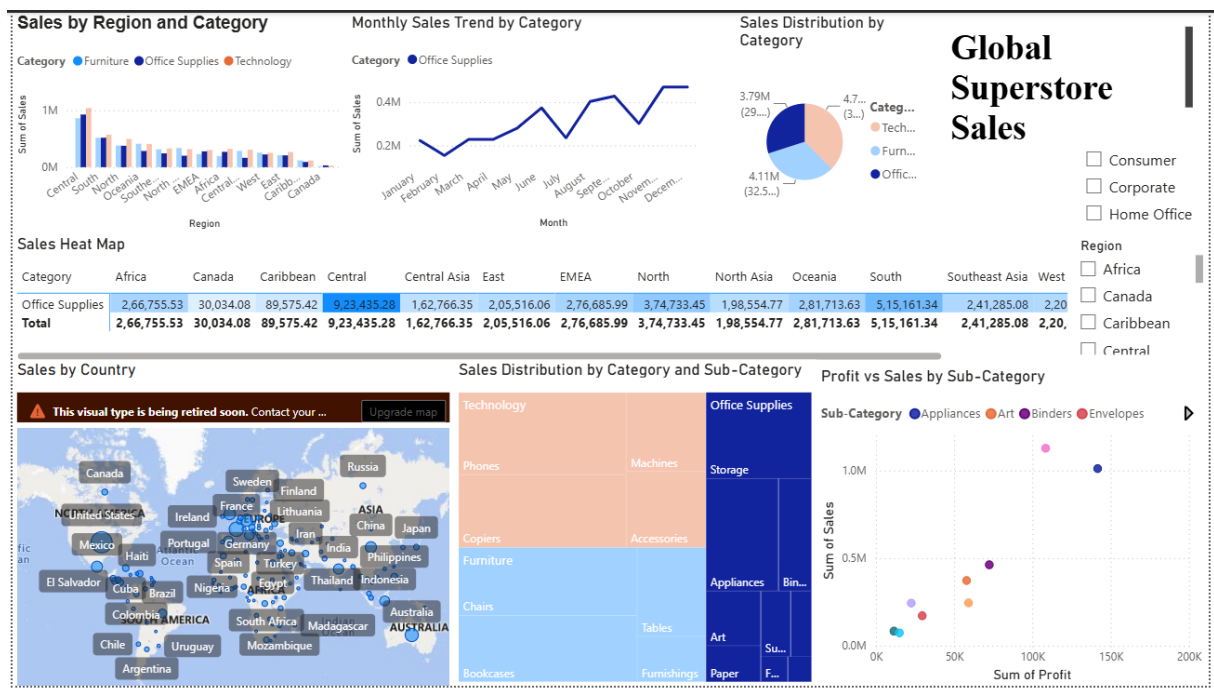
Add interactive filters and tooltips for dynamic exploration.



FINAL DASHBOARD : (TABLEAU DASHBOARD)



POWER BI DASHBOARD :



POWER BI DASHBOARD LINK :

<https://kumaragurudtsteam->

my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/English/BI%20LAB%20EX%202.pbix?csf=1&web=1&e=92C80U

GITHUB LINK :

Experiment Dataset:

Global Superstore Sales Dataset

Source: <https://www.kaggle.com/apoorvaappz/global-super-store-dataset>

Fields Used:

Order Date

Region

Category

Sub-Category

Sales

Profit

Country

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart:

1.What regions had the highest and lowest total sales?

The **Central** region recorded the **highest total sales**, while **Canada** recorded the **lowest total sales**.

2.How does the performance of categories vary across regions?

Technology generated the highest sales in most regions, followed by Furniture and Office Supplies. The Central region performed exceptionally well across all categories, whereas Canada and the Caribbean showed comparatively lower sales

Line Chart:

1.Identify the month with peak sales.

The highest sales were recorded in **November**, with sales reaching approximately **600K**.

2.Which category shows consistent growth in sales over time?

The **Technology** category showed the most consistent growth throughout the year, maintaining higher sales than the other categories.

Pie Chart:

1.Which category contributes the most to overall sales?

The **Technology** category contributed the highest share of overall sales with approximately **4,744,557**, followed by **Furniture (4,110,874)** and **Office Supplies (3,787,070)**.

2. Are there any categories with negligible sales contributions?

No. All three categories contributed significantly to the total sales, although **Office Supplies** had the smallest share among them.

Scatter Plot:

1. Is there a noticeable correlation between profit and sales?

Yes. The scatter plot indicates a **positive correlation** between sales and profit, as products with higher profits generally have higher sales.

2. Which sub-category shows the highest sales but low profit?

Storage shows relatively high sales with lower profit compared to **Phones** and **Chairs**. Some sub-categories also have negative profit despite generating sales.

Heatmap:

1. Which region-category combination has the highest sales?

The **Central Region – Technology** combination recorded the highest sales, with approximately **1,038,450**.

2. Identify any regions with poor sales performance across all categories.

Canada recorded the lowest sales across all three categories, followed by the **Caribbean** region.

Geographical Map:

1. Which country has the highest total sales?

The **United States** recorded the highest total sales, represented by the largest and darkest circle on the map.

2. Are there regions with minimal or no sales?

Yes. Countries such as **Argentina**, **South Africa**, and **New Zealand** recorded comparatively lower sales than major markets.

Tree Map:

1. Which sub-category dominates the sales in each category?

- **Technology:** Phones
- **Furniture:** Chairs
- **Office Supplies:** Binders

These sub-categories occupy the largest areas in the treemap.

2.Are there any sub-categories with unexpectedly low sales?

Yes. Sub-categories such as **Fasteners, Labels, and Supplies** occupy much smaller areas, indicating lower sales.

Dashboard:

1.What insights can you derive when combining all visualizations?

The dashboard gives a complete overview of sales performance across regions, categories, countries, and time. It reveals that Technology is the leading category, sales peak during November, the United States contributes the highest sales, and Central is the best-performing region. It also highlights the relationship between sales and profit and identifies strong and weak performing sub-categories.

2.How do filters enhance the utility of the dashboard?

Filters enable users to interactively analyze the data by Region, Category, Year, Customer Segment, and other dimensions. They simplify comparisons, support focused analysis, and help users identify trends and business insights more effectively.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		